Data-Driven Modeling and Optimization of a Gas-Steam Combined Cycle

Yichen Wang

Xingjian College, Tsinghua University, Beijing 100084, China

Introduction

This study builds a data-driven thermodynamic analysis framework for a gas-steam combined cycle unit, covering working-fluid modeling, gas-turbine analysis, steam/water-side analysis, and whole-plant efficiency-factor analysis. Historical operation data are used to reconstruct key thermodynamic states, quantify energy distribution, and compare power-only and heating modes, providing a basis for operation optimization and long-term performance assessment.

1. Working-Fluid Modeling

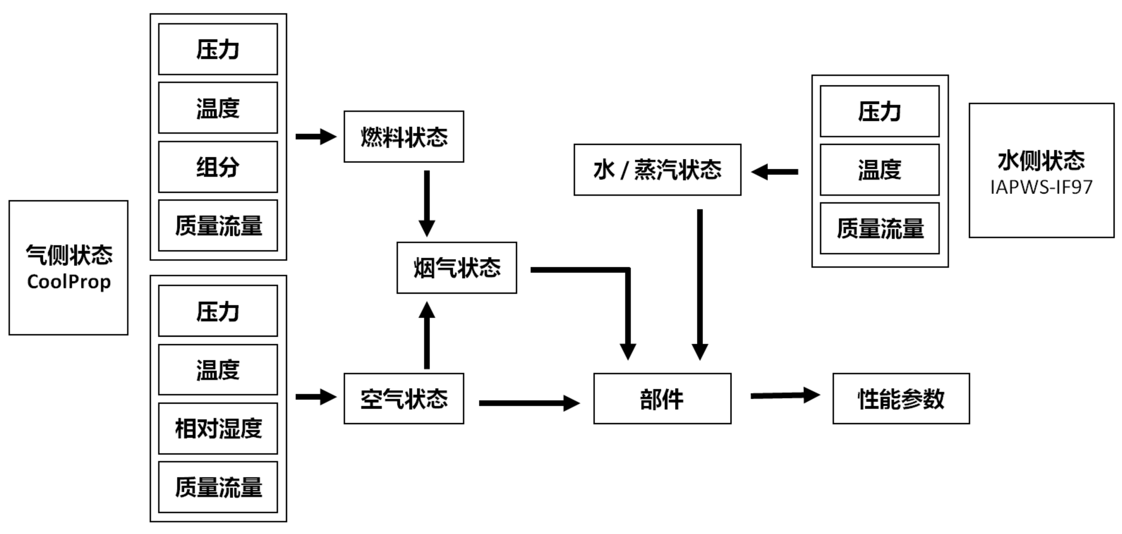

Fig. 1. Working-fluid state and property modeling.

Gas, air, fuel, water, and steam streams are represented by unified flow-state objects. Thermodynamic properties such as enthalpy, entropy, energy flow, and exergy flow are calculated consistently for downstream component models.

2. Gas-Turbine Analysis

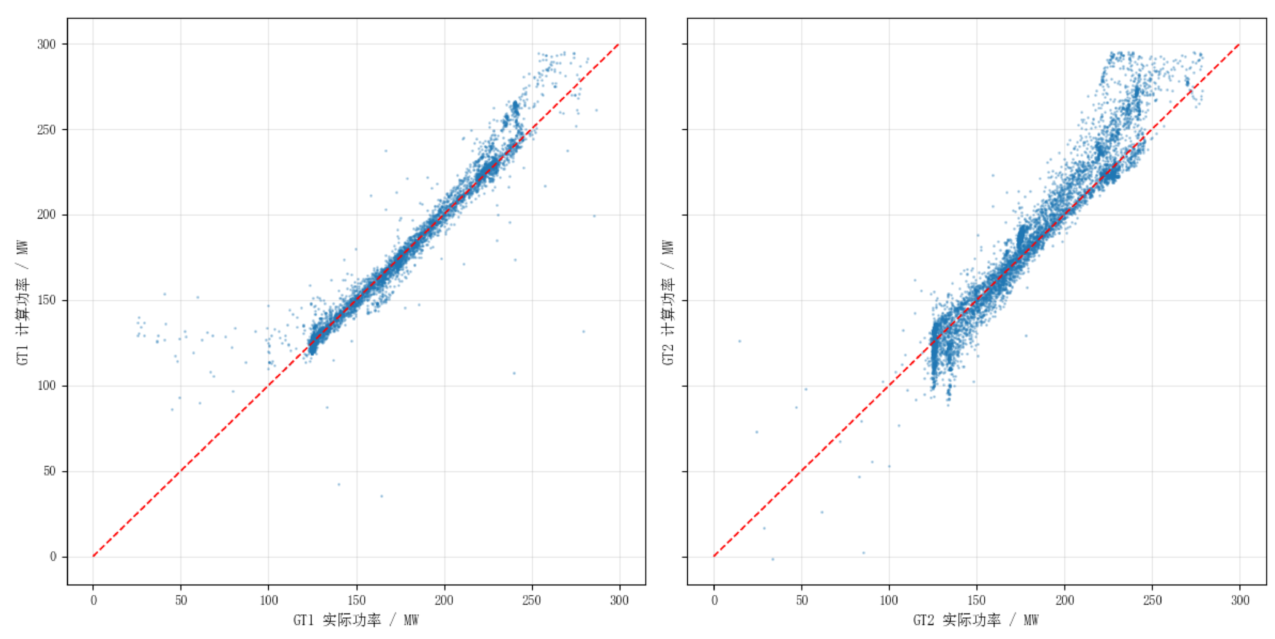

Fig. 2. Gas-turbine calculated and measured power.

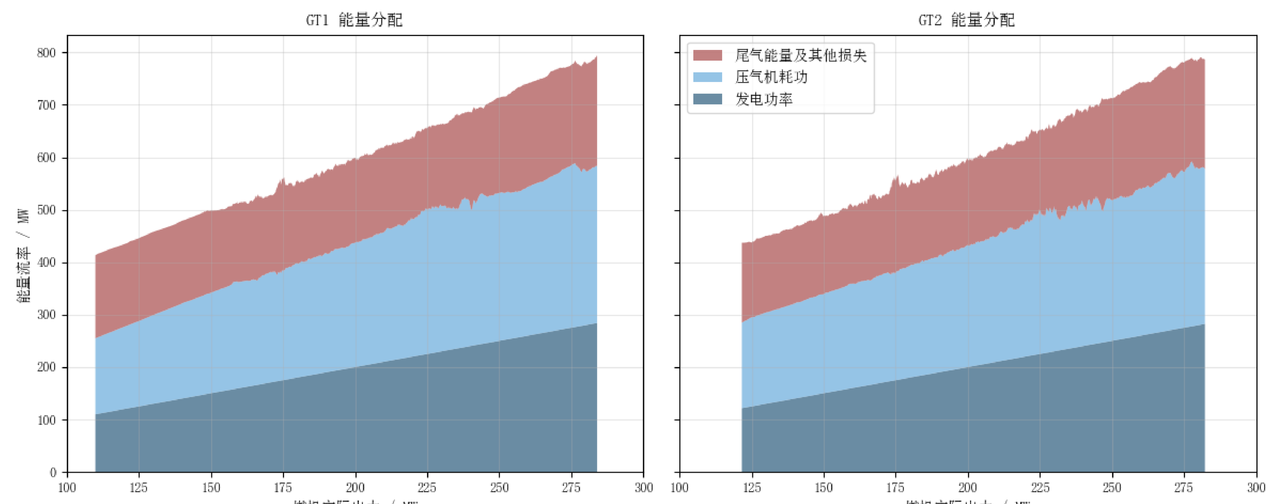

Fig. 3. Fuel-energy distribution in the gas turbine.

The gas-turbine model evaluates compressor work, turbine work, net power, fuel energy, and exhaust heat. The energy distribution clarifies how fuel input is divided into electric output, internal compression work, and recoverable exhaust energy.

3. Steam/Water-Side Analysis

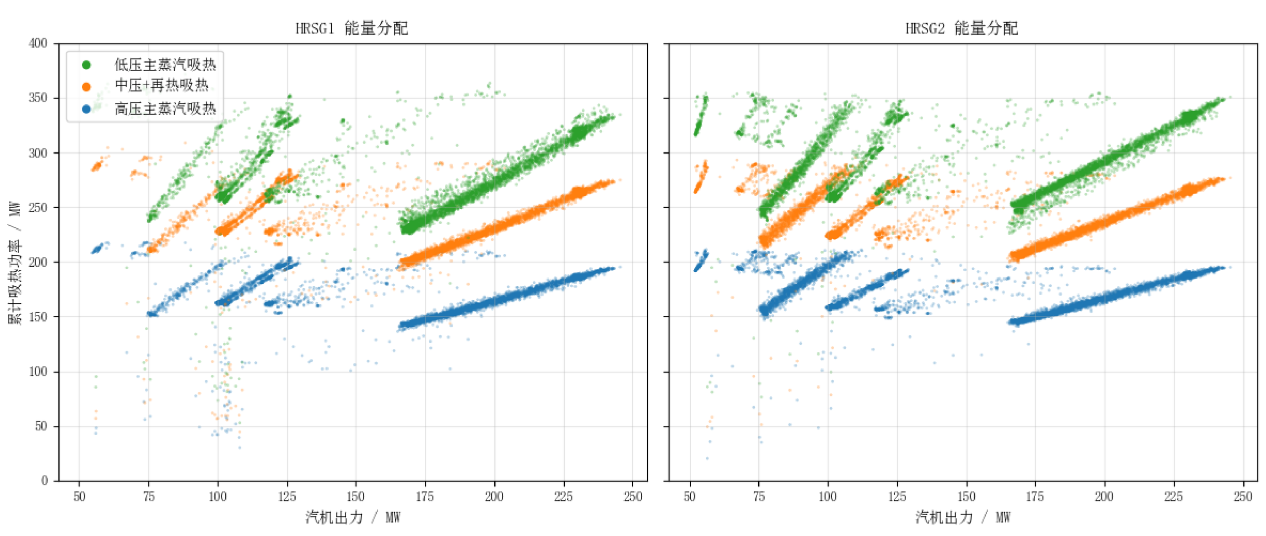

Fig. 4. Heat absorption distribution on the steam/water side.

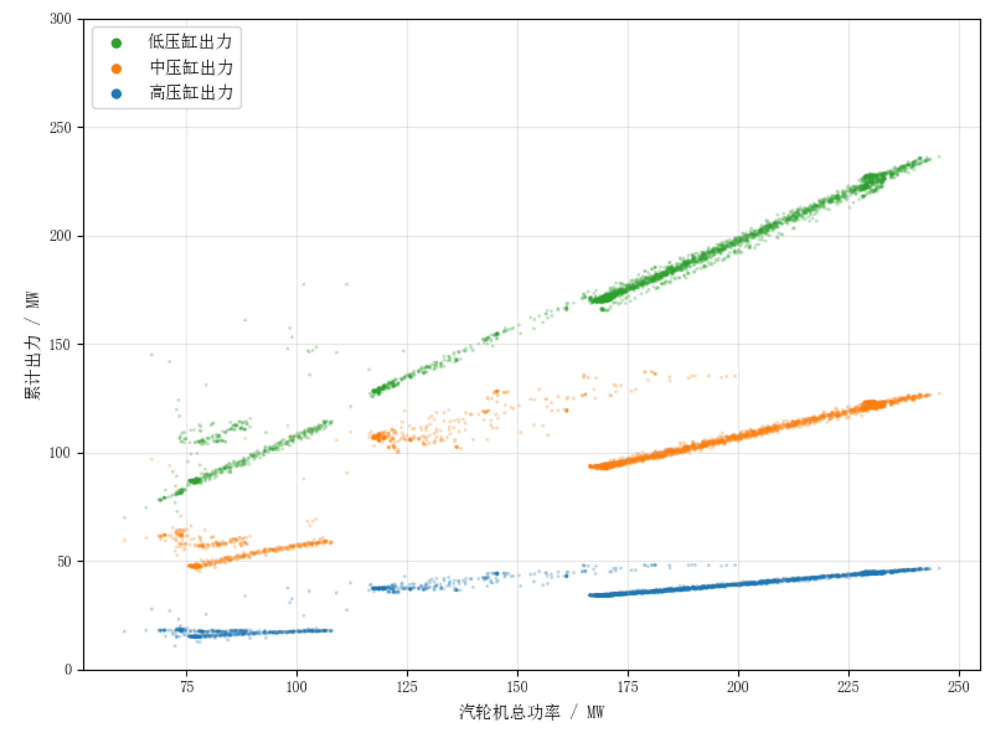

Fig. 5. Power distribution among HP, IP, and LP cylinders.

The HRSG transfers gas-side heat to high-pressure steam, intermediate-pressure/reheat steam, and low-pressure steam. The steam turbine further converts steam energy into cylinder power, while heating extraction changes the effective output definition.

4. Whole-Plant Efficiency

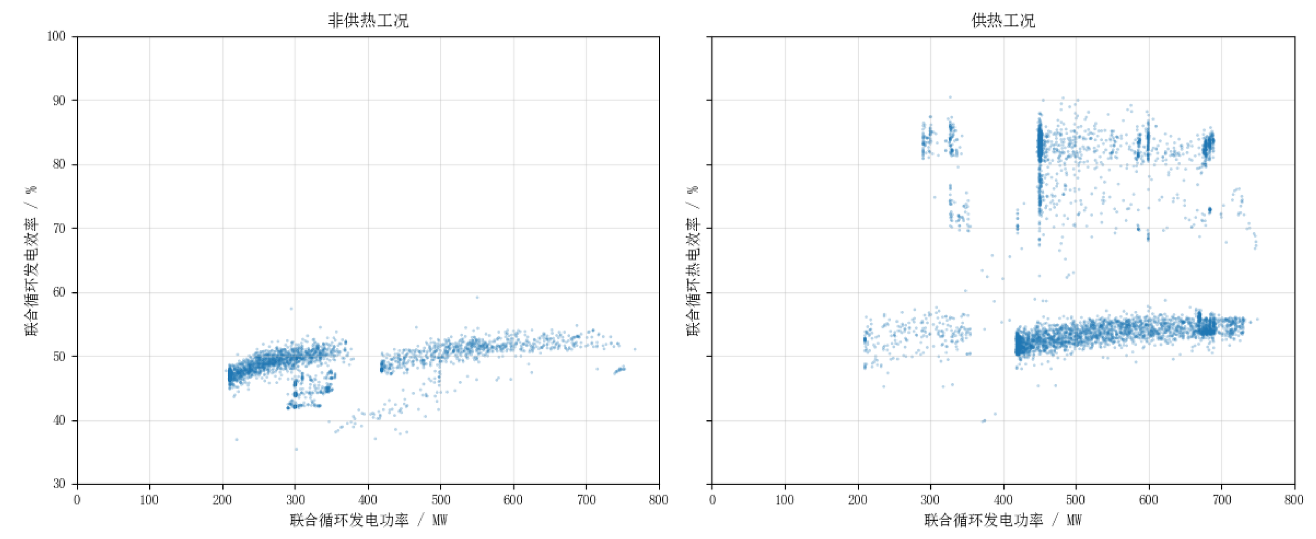

Fig. 6. Combined-cycle efficiency versus total plant power.

The power-only samples form separated clusters because the plant switches between single- and dual-gas-turbine operation. Heating samples show higher overall utilization since heat supply is counted as useful output in the CHP efficiency definition.

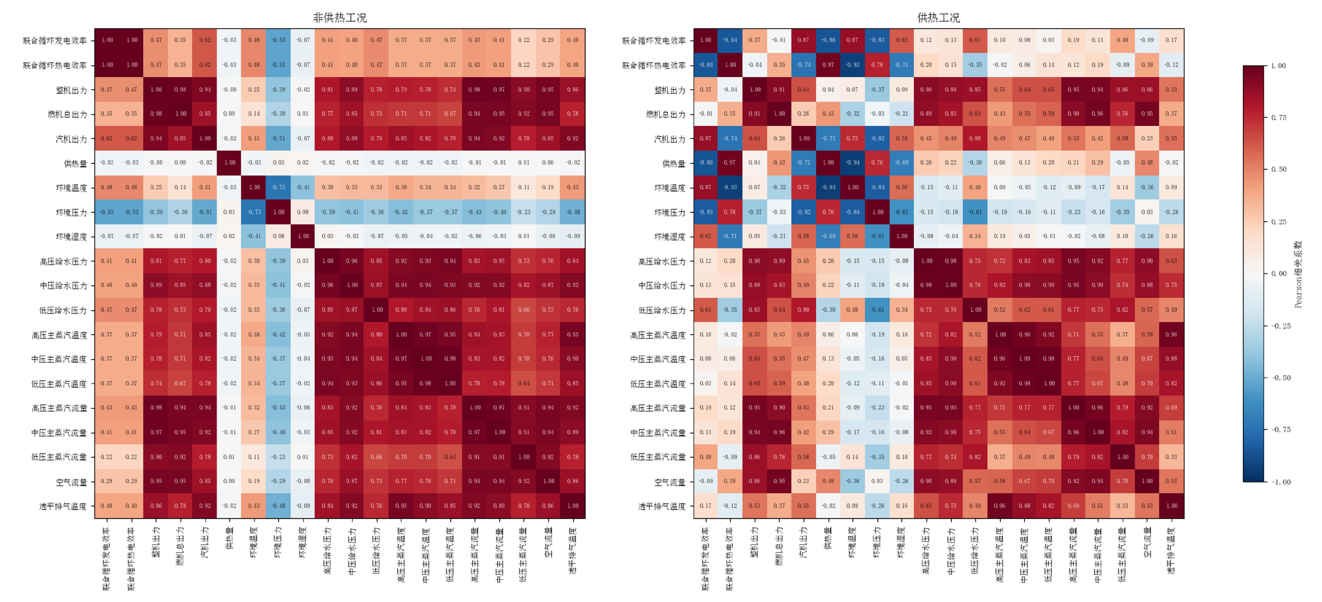

Fig. 7. Pearson correlation matrices for power-only and heating modes.

Power-only operation is evaluated by electric efficiency, while heating operation is evaluated by combined heat-and-power efficiency. The two operating modes are analyzed separately because heat supply changes the numerator of the efficiency definition.

5. Optimization Methods

The correlation and scatter analyses suggest four practical optimization directions: operating-mode classification, heat-supply scheduling, steam-side parameter control, and gas-turbine air-flow management.

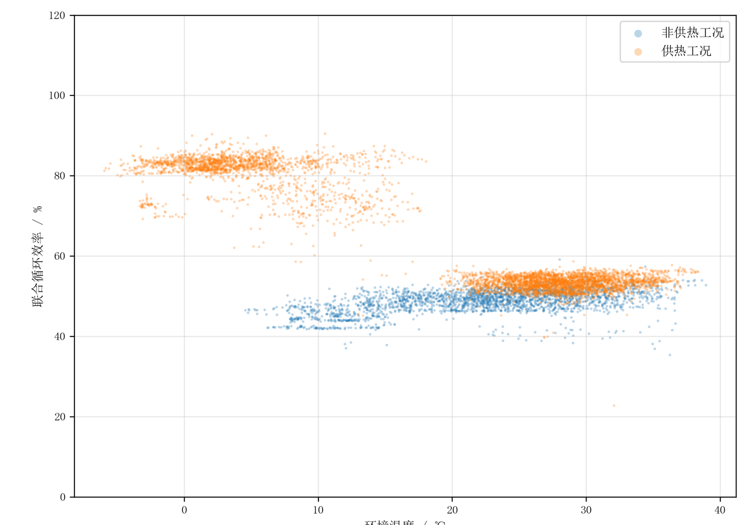

Fig. 8. Ambient temperature versus efficiency.

Fig. 9. Heat supply versus efficiency.

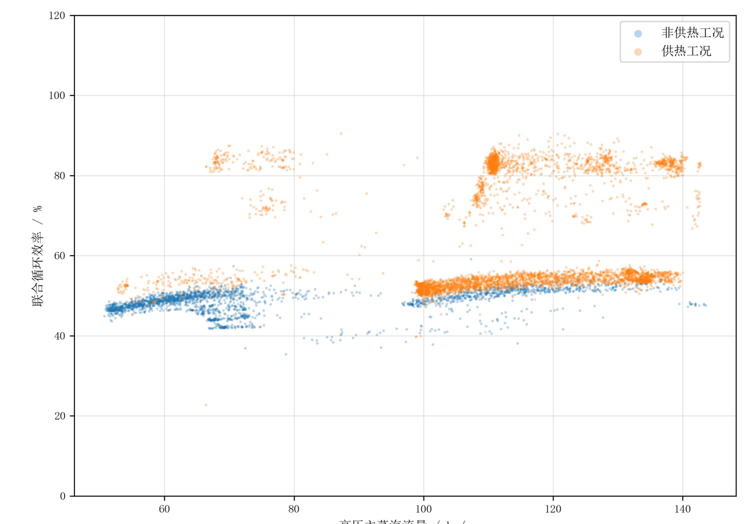

Fig. 10. HP main-steam flow versus efficiency.

Fig. 11. Air flow versus efficiency.

6. Summary and Future Work

- A plant-wide calculation workflow was established from working-fluid states to component and whole-plant performance.
- Gas-turbine analysis identifies the distribution of fuel energy among power output, compressor work, and exhaust heat.
- Steam/water-side analysis links HRSG heat absorption with steam-turbine cylinder power and heating extraction.
- Whole-plant analysis shows that power-only and heating modes must be evaluated using different efficiency definitions.
- Future work will combine operating-mode classification with long-term degradation assessment.

Data source: raw operation data collected from Beijing Thermal Power Plant.